STAT 516: Homework 6 Answer Key

1. Let (X, Y) have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 4xy, & 0 < x < 1,\ 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find the marginal PDFs $f_X(x)$ and $f_Y(y)$.

(b) Determine whether X and Y are independent.

Solution.

For $0 < x < 1$,

$$f_X(x) = \int_0^1 f_{X,Y}(x,y)\,dy = \int_0^1 4xy\,dy.$$

Therefore,

$$f_X(x) = 4x\left[\frac{y^2}{2}\right]_0^1 = 2x.$$

Thus,

$$f_X(x) = \begin{cases} 2x, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Similarly, for $0 < y < 1$,

$$f_Y(y) = \int_0^1 f_{X,Y}(x,y)\,dx = \int_0^1 4xy\,dx.$$

Therefore,

$$f_Y(y) = 4y\left[\frac{x^2}{2}\right]_0^1 = 2y.$$

Thus,

$$f_Y(y) = \begin{cases} 2y, & 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

For $0 < x < 1$ and $0 < y < 1$,

$$f_X(x)f_Y(y) = (2x)(2y) = 4xy = f_{X,Y}(x,y).$$

Since the joint support is rectangular and

$$f_{X,Y}(x,y) = f_X(x)f_Y(y),$$

X and Y are independent.

2. Let (X, Y) have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} \frac{6}{5}(x^2 + y), & 0 < x < 1,\ 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate

$$\mathrm{E}[2X - Y + XY].$$

Solution.

Using the expectation formula for a function of two random variables,

$$\mathrm{E}[2X - Y + XY] = \int_0^1 \int_0^1 (2x - y + xy)\frac{6}{5}(x^2 + y)\, dy\, dx.$$

Expanding the integrand,

$$(2x - y + xy)(x^2 + y) = 2x^3 - x^2y + x^3y + 2xy - y^2 + xy^2.$$

Therefore,

$$\mathrm{E}[2X - Y + XY] = \frac{6}{5}\int_0^1 \int_0^1 \left(2x^3 - x^2y + x^3y + 2xy - y^2 + xy^2\right)\, dy\, dx.$$

Integrating term by term,

$$\mathrm{E}[2X - Y + XY] = \frac{6}{5}\left(\frac{1}{2} - \frac{1}{6} + \frac{1}{8} + \frac{1}{2} - \frac{1}{3} + \frac{1}{6}\right).$$

Thus,

$$\mathrm{E}[2X - Y + XY] = \frac{6}{5} \cdot \frac{19}{24} = \frac{19}{20}.$$

3. Let X represent the age of an insured automobile involved in an accident. Let Y represent the length of time the owner has insured the automobile at the time of the accident.

Suppose X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{64}(10 - xy^2), & 2 \le x \le 10,\ 0 \le y \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate

$$\mathrm{E}[X].$$

Solution.

By definition,

$$\mathrm{E}[X] = \int_2^{10} \int_0^1 x f_{X,Y}(x,y)\, dy\, dx.$$

Therefore,

$$\mathrm{E}[X] = \int_2^{10} \int_0^1 x \cdot \frac{1}{64}(10 - xy^2)\, dy\, dx.$$

Integrating first with respect to y,

$$\mathrm{E}[X] = \frac{1}{64} \int_2^{10} x \left(10 - \frac{x}{3}\right) dx.$$

Thus,

$$\mathrm{E}[X] = \frac{1}{64} \int_2^{10} \left(10x - \frac{x^2}{3}\right) dx.$$

Hence,

$$\mathrm{E}[X] = \frac{1}{64} \left[5x^2 - \frac{x^3}{9}\right]_2^{10}.$$

Evaluating,

$$\mathrm{E}[X] = \frac{1}{64} \left(\frac{3500}{9} - \frac{172}{9}\right) = \frac{1}{64} \cdot \frac{3328}{9} = \frac{52}{9} \approx 5.7778.$$

4. Let X and Y be continuous random variables with joint PDF

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{4}, & 0 \le x \le 2,\ x - 2 \le y \le x, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate

$$\mathrm{E}[X^3 Y].$$

Solution.

By definition,

$$\mathrm{E}[X^3 Y] = \int_0^2 \int_{x-2}^{x} x^3 y \cdot \frac{1}{4}\, dy\, dx.$$

Therefore,

$$\mathrm{E}[X^3 Y] = \frac{1}{4} \int_0^2 x^3 \left[\frac{y^2}{2}\right]_{x-2}^{x} dx.$$

Since

$$x^2 - (x-2)^2 = x^2 - (x^2 - 4x + 4) = 4x - 4,$$

we obtain

$$E[X^3Y] = \frac{1}{8}\int_0^2 x^3(4x-4)\,dx.$$

Thus,

$$E[X^3Y] = \frac{1}{2}\int_0^2 (x^4-x^3)\,dx.$$

Therefore,

$$E[X^3Y] = \frac{1}{2}\left[\frac{x^5}{5} - \frac{x^4}{4}\right]_0^2.$$

Hence,

$$E[X^3Y] = \frac{1}{2}\left(\frac{32}{5} - 4\right) = \frac{1}{2}\cdot\frac{12}{5} = \frac{6}{5}.$$

5. The profit of a new product is given by

$$Z = 3X - Y - 5.$$

Assume X and Y are independent random variables with

$$\mathrm{Var}(X) = 1 \qquad \text{and} \qquad \mathrm{Var}(Y) = 2.$$

Calculate

$$\mathrm{Var}(Z).$$

Solution.

The constant -5 does not affect the variance. Therefore,

$$\mathrm{Var}(Z) = \mathrm{Var}(3X - Y).$$

Since X and Y are independent,

$$\mathrm{Cov}(X, Y) = 0.$$

Thus,

$$\mathrm{Var}(Z) = 3^2\,\mathrm{Var}(X) + (-1)^2\,\mathrm{Var}(Y).$$

Therefore,

$$\mathrm{Var}(Z) = 9(1) + 1(2) = 11.$$

6. The profit of a new product is given by

$$Z = 5X - 2Y - 3.$$

Assume

$$\mathrm{Var}(X) = 1, \qquad \mathrm{Var}(Y) = 2, \qquad \mathrm{Cov}(X, Y) = -1.$$

Calculate

$$\mathrm{Var}(Z).$$

Solution.

Using

$$\mathrm{Var}(aX + bY) = a^2\,\mathrm{Var}(X) + b^2\,\mathrm{Var}(Y) + 2ab\,\mathrm{Cov}(X, Y),$$

we obtain

$$\mathrm{Var}(Z) = 5^2\,\mathrm{Var}(X) + (-2)^2\,\mathrm{Var}(Y) + 2(5)(-2)\,\mathrm{Cov}(X, Y).$$

Substituting the given values,

$$\mathrm{Var}(Z) = 25(1) + 4(2) + 2(5)(-2)(-1) = 53.$$

7. An actuary analyzes a company's annual personal auto claim, M, and annual commercial auto claim, N. Suppose

$$\mathrm{Var}(M) = 1600, \qquad \mathrm{Var}(N) = 900,$$

and the correlation between M and N is

$$\rho(M, N) = 0.64.$$

Calculate

$$\mathrm{Var}(M + N).$$

Solution.

Using the definition of correlation,

$$\mathrm{Corr}(M, N) = \frac{\mathrm{Cov}(M, N)}{\sqrt{\mathrm{Var}(M)}\sqrt{\mathrm{Var}(N)}}.$$

Therefore,

$$\mathrm{Cov}(M, N) = \mathrm{Corr}(M, N)\sqrt{\mathrm{Var}(M)}\sqrt{\mathrm{Var}(N)}.$$

Substituting the given values,

$$\mathrm{Cov}(M, N) = 0.64\sqrt{1600}\sqrt{900} = 0.64(40)(30) = 768..$$

Now,

$$\mathrm{Var}(M + N) = \mathrm{Var}(M) + \mathrm{Var}(N) + 2\,\mathrm{Cov}(M, N).$$

Thus,

$$\mathrm{Var}(M + N) = 1600 + 900 + 2(768) = 4036.$$

8. Let X and Y be continuous random variables with joint PDF

$$f_{X,Y}(x,y) = \begin{cases} \frac{8}{3}xy, & 0 \le x \le 1,\ x \le y \le 2x, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate

$$\operatorname{Cov}(X,Y).$$

Solution.

We use

$$\operatorname{Cov}(X,Y) = \mathrm{E}[XY] - \mathrm{E}[X]\,\mathrm{E}[Y].$$

First,

$$\mathrm{E}[X] = \int_0^1 \int_x^{2x} x \cdot \frac{8}{3}xy \, dy \, dx.$$

Thus,

$$\mathrm{E}[X] = \frac{8}{3}\int_0^1 x^2 \left[\frac{y^2}{2}\right]_x^{2x} dx = \frac{8}{3}\int_0^1 x^2 \cdot \frac{3x^2}{2} \, dx = 4\int_0^1 x^4 \, dx = \frac{4}{5}.$$

Next,

$$\mathrm{E}[Y] = \int_0^1 \int_x^{2x} y \cdot \frac{8}{3}xy \, dy \, dx.$$

Thus,

$$\mathrm{E}[Y] = \frac{8}{3}\int_0^1 x \left[\frac{y^3}{3}\right]_x^{2x} dx = \frac{56}{9}\int_0^1 x^4 \, dx = \frac{56}{45}.$$

Now calculate

$$\mathrm{E}[XY] = \int_0^1 \int_x^{2x} xy \cdot \frac{8}{3}xy \, dy \, dx = \frac{8}{3}\int_0^1 x^2 \cdot \frac{7x^3}{3} \, dx = \frac{56}{9}\int_0^1 x^5 \, dx = \frac{56}{9} \cdot \frac{1}{6} = \frac{28}{27}.$$

Finally,

$$\operatorname{Cov}(X,Y) = \frac{28}{27} - \left(\frac{4}{5}\right)\left(\frac{56}{45}\right) = \frac{28}{675} \approx 0.04148.$$